Social Media Sentiment Analysis Report: Lawrence Technological University

Course: INT6303 – Social Media Data Analytics
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1. INTRODUCTION

1.1 General Domain

Sentiment analysis is a subfield of Natural discourse Processing (NLP), a larger science that uses computational techniques to interpret and comprehend human discourse. In the context of higher education, where institutions get vast amounts of user-generated content from alumni, students, teachers, and the general public on sites like Glassdoor, Reddit, and Twitter, this study investigates sentiment analysis. Public opinion can be greatly influenced by these platforms, which act as unofficial review systems.

1.2 Importance of Sentiment Analysis

Universities may proactively monitor comments, identify reputational threats, and pinpoint opportunities for development by having a thorough understanding of sentiment. This can have an impact on a number of operational areas, such as faculty involvement, marketing strategy, admissions outreach, and even curriculum modification. Since social media is the main platform for student voices, it is crucial for institutional development and transparency to be able to monitor, evaluate, and understand their opinions.

2. PROBLEM STATEMENT & DESCRIPTION

2.1 Problem Definition

Automated solutions for analyzing vast amounts of unstructured feedback data from social media sites are lacking at universities. Their capacity to intervene promptly is hampered by this. For instance, without adequate monitoring tools, a rise in unfavorable attitude regarding campus safety or administrative delays can go missed.

2.2 Expected Functionalities

An efficient system should:

- Preprocess and absorb user-generated content;
- Use a strong algorithm to determine sentiment polarity;
- Examine patterns over time and across platforms;
- Find reoccurring issues and promising aspects;
- Display the analysis with graphs and tables.

3. SOLUTION OVERVIEW

3.1 Initial Specifications

Dataset Requirements:

The dataset needs to be based on or replicate actual user posts that mention the university. Metadata such the date, platform, sentiment label, and text content should be included in every entry. All sentiment classifications should ideally be fairly represented in the dataset.

Data Analysis Techniques:

- Text cleaning (lemmatization, lowercase, and stopword removal);
- TextBlob for polarity scoring;
- Token and n-gram models for frequency analysis;
- Matplotlib and Seaborn for visualization.
- Using Pandas for grouping and trend detection

3.2 Scope of the Project

Included:

- Creating a unique 500-row dataset for LTU with equal sentiment categories;
- Using TextBlob for rule-based sentiment classification;
- Visualizing monthly trends and sentiment distribution; and
- Extracting sentiment-based top words and phrases.

Excluded:

- Real-time data collecting by streaming through Reddit or Twitter APIs
- Sentiment analysis based on neural networks (like BERT)
- Features for user interfaces or interactive dashboards

4. DATA OVERVIEW

4.1 Dataset – Key Features

Five hundred carefully created entries that mimic posts from Glassdoor, Reddit, and Twitter make up the dataset. Fields for Username, Date, Platform, UserType, PostType, Text, Hashtags, and Sentiment are all included in each entry. The TextBlob polarity scoring approach was also used to create a derived column for anticipated sentiment.

4.2 Data Source

The synthetic dataset is intended to replicate real-world content found in mentions of public universities on specific social media platforms. It displays typical language usage patterns among employees and students.

4.3 Dataset Size and Scope

- 500 rows of data with labels;
- Text lengths range from 15 to 50 words;
- Positive, negative, and neutral class distribution is balanced;
- Between January and December of 2024

5. ANALYSIS AND FINDINGS

5.1 Sentiment Distribution

45.8% of the posts in our sample were positive, 229 posts were negative, 215 posts were negative, and 56 posts were neutral. People are more likely to voice strong opinions, whether they are favorable or negative, in the actual world, which is reflected in this tiny imbalance. Despite being manually created, the dataset replicates authentic user behavior patterns found in real-world sentiment data from social media.

5.2 Monthly Trends

Month-by-month sentiment analysis showed variations consistent with believable academic occurrences. Negative attitude increased in March, which was frequently ascribed to administrative problems, midterm pressures, or discontent with the cold weather. December had the most neutral posts, indicating that students were finishing up their exams and thinking back on the semester. The most upbeat posts were found in January and November, which probably reflected the excitement surrounding the start of new semesters and school activities.

5.3 Word Frequency

After the text was cleaned and preprocessed, a token-level frequency analysis was carried out. Positive terms that were used the most frequently were "offers," "great," "experience," "excellent," and "supportive." These terms were commonly used in expressions that praised university professors and resources.

The following terms were frequently used to describe unfavorable sentiment: "outdated," "unresponsive," "frustrating," "poor," and "communication." These point to recurrent themes of discontent with administrative procedures and infrastructure.

5.4 Phrase Analysis

Frequent phrases were identified by extracting bigrams (2-grams). Positive 2-grams that stood out the most were "great experience," "excellent professors," and "offers resources." These show recurrent displays of contentment with the caliber of education and institutional assistance. Negative 2-grams, on the other hand, included "not recommended," "outdated systems," and "communication not," all of which indicate poor support and user experience systems.

5.5 Visualization Insights

An essential component of interpretation was the use of visualizations. Line charts showed the distribution of emotion over the months, emphasizing both increases and decreases. Top words in each sentiment class were displayed in bar graphs, and monthly counts by sentiment were compiled in tables. These clarified the general tendencies and supported the textual findings. Additionally, word clouds—where font size indicated word frequency—were utilized for instant visual reference.

5.6 Cross-Verification

The sentiment predictions made by the model with TextBlob were contrasted with entries that were manually tagged. Disparities were brought to light by this verification method, particularly when sarcasm, conflicting viewpoints, or implied sentiment were present. Although TextBlob performed well when handling simple sentiment, it frequently had trouble with complex or context-sensitive language, which suggests that more sophisticated models could be used to improve it.

6. CONCLUSION

This project shows how sentiment analysis in an academic setting can yield useful insights from social media data. We were able to create a pipeline that replicates sentiment monitoring systems in the real world by using a synthetically structured dataset and simple natural language processing technologies. In the end, the findings improve institutional openness and strategy by assisting institutions in recognizing and addressing significant trends in student feedback.

6.1 Future Work

- For contextual analysis, use BERT or RoBERTa instead of TextBlob.
- Create a live dashboard for continued institutional usage;
- Analyze sentiment by user type (faculty, alumni, and students); and
- Expand the dataset to incorporate real-time API data and news items.

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